



魅力统计

Fascinating Statistics

第二章 认识概率

Chapter 2 Understanding Probability

Chapter 2 Understanding Probability

- **2.1 Determinism and Randomness**
- 2.2 Development of Probability Theory
- 2.3 Basic Concepts of Probability Theory
- 2.4 Calculating Probability
- 2.5 Conditional Probability and Independence



2.1 Determinism and Randomness



SOLVAY CONFERENCE 1927

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2.1 Determinism and Randomness

- The Fifth Solvay Conference is often regarded as the decisive showdown between Einstein and Bohr on their respective views of quantum theory. It was also from this event that Einstein's famous assertion, "God does not play dice" became widely known.
- A core idea of quantum mechanics is that everything in the quantum world is probabilistic (e.g., Heisenberg's Uncertainty Principle and Schrödinger's cat).
- The general public believes that Einstein was convinced the universe operated in a classical physics manner, ticking away mechanically like a clock, with each moment determining the next.
- However, after in-depth analysis of Einstein's words by many historians, philosophers, and physicists, it is found that he was not opposed to quantum mechanics or randomness itself.
- Actually, Einstein acknowledged the randomness in quantum phenomena, as early as 1916, he demonstrated that when an atom emits a photon, the timing and angle of emission are random.





2.1 Determinism and Randomness

- Einstein believed that the randomness we currently attribute to quantum behavior is simply due to our failure to discover the "hidden variables" or some yet unknown theories.
- Researchers use probability to explain quantum behavior, in his view, stems from insufficient knowledge rather than the inherent randomness of the universe.
- Just like rolling a dice, although the outcome of each throw appears random, if we knew the initial position, direction, wind speed, and all other environmental factors, the final result of the dice roll should, in theory, be deterministic.
- Whether the universe operates like a clockwork mechanism or a dice-rolling table is beyond the scope of this course, and even today, humanity cannot claim a complete understanding of the quantum mechanics.
- Before the era of scientists like Bohr and Einstein, the idea of randomness in physical laws was scarcely believed, and the use of statistical methods was rare.



2.1 Determinism and Randomness

- With the advancement of modern statistics, not only in physics but across various fields, the concept of randomness has gradually been accepted.
- Statistics, as a powerful tool, has solved an increasing number of problems, making the question "Does God play dice?" less significant.
- Let's end this section with a quote from C.R. Rao, a famous statistician,
 - All knowledge is, in final analysis, history. 在终极的分析中，一切知识都是历史。
 - All sciences are, in the abstract, mathematics. 在抽象的意义下，一切科学都是数学。
 - All judgements are, in their rationale, statistics. 在理性的基础上，一切判断都是统计学。



Chapter 2 Understanding Probability

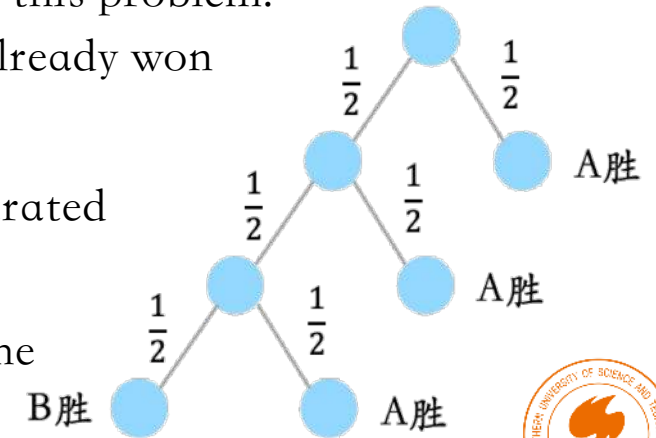
- 2.1 Determinism and Randomness
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2.2 Development of Probability Theory



- A landmark event in the history of probability theory is the [problem of points](#).
 - First proposed by [Luca Pacioli](#), known as the father of modern accounting, in his book *Summary of arithmetic, geometry, proportions and proportionality* published in 1494.
 - Two gamblers A and B both have a winning rate of 50% when playing a game. They agree that the one who wins 6 rounds first will take all the stakes. When A wins 5 rounds and B wins 3 rounds, the game is stopped for some reason. How should the stakes be allocated to A and B?
 - The answer Pacioli gave at the time was based on the number of games that A and B had won, i.e., 5:3.
 - In 1539, the renowned Renaissance scholar [Cardano](#) also attempted to solve this problem. Although his solution was incorrect, his idea of focusing not on the games already won but on the future games was groundbreaking.
 - In 1654, [Pascal](#) sought to tackle this problem, and his approach can be illustrated using the tree-like structure shown in the diagram on the right.
 - Pascal thought that A should get $\frac{1}{2} + \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{7}{8}$ of the stakes, so the answer should be 7:1.



2.2 Development of Probability Theory



- A landmark event in the history of probability theory is the **problem of points**.
 - At the time, Pascal was unsure whether his calculation was correct, so he wrote to his friend **Fermat**.
 - Fermat proposed another approach: the game would end in at most three rounds and there would be a total of eight possible outcomes, AAA, AAB, ABA, ABB, BAA, BAB, BBA, and BBB. Only in the case of BBB would player B win the stake, making the odds 7:1, which was correct.
- Although the problem of points was simple, it holds an important place in the history of scientific development.
- In the correspondence between Pascal and Fermat, many other probability problems were also discussed, and Pascal provided clear and comprehensive solutions.
- As a result, later generations have credited Pascal with formally establishing probability theory in **1654**, with Fermat being a significant contributor.



2.2 Development of Probability Theory

- At that time, people could not imagine the immense power that “probability” would bring. However, the door to “randomness” that would change the world had been opened.
- Then probability theory continued to develop. Although a mathematical axiomatic system had not yet been established, it already possessed a solid mathematical foundation, enabling it to solve many practical problems.
- The probability theory in this period is known as the **classical probability theory** (古典概率论).
- It is generally considered that **Pierre Laplace's *Analytical Theory of Probability***, published in 1812, marked the refinement of classical probability theory.
- Classical probability theory, based on probability calculations and the law of large numbers, focused on solving practical problems.

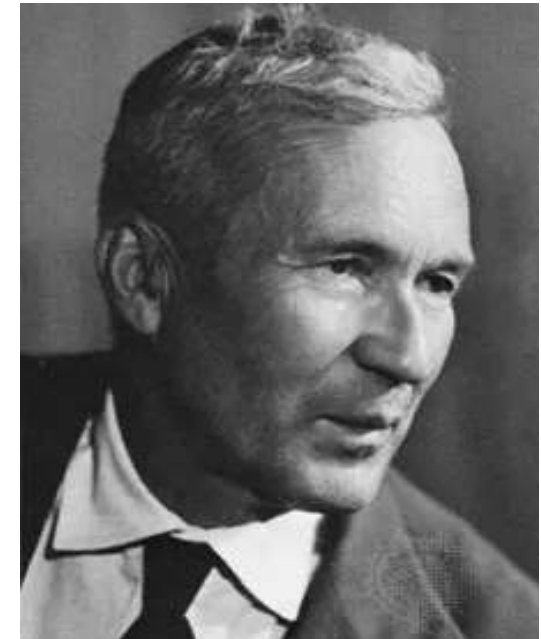
Pierre Laplace (1749-1827)



2.2 Development of Probability Theory

- In 1933, the Soviet mathematician **Andrey Kolmogorov** combined sample spaces with measure theory to establish the axiomatic system of probability theory.
- The publication of his work *Foundations of the Theory of Probability* also became a landmark event in the birth of **modern probability theory** (现代概率论).
- However, for most practical applications, there is no fundamental difference between classical and modern probability theory.
- Therefore, even without a profound mathematical background, one can still use probability theory as a powerful tool.
- Probability theory is defined as a mathematical discipline that studies the statistical regularity of random phenomena.
- Its primary subjects include random experiments, random events, random variables, probability distributions, etc.

Andrey Kolmogorov
(1903-1987)



2.2 Development of Probability Theory

- Before delving into these topics, we should understand the conceptual framework that probability theory uses to solve problems.
- At its core, probability theory transforms local randomness into global certainty.
- Probability theory allows us to make mathematically deterministic judgments about random events.
 - In the problem of points, we cannot know for sure whether A or B will win, but how to allocate the stakes is undeniably clear.
 - For Schrödinger's cat, although we do not know whether it will be alive or dead in the next second, the superposition of its possible states of life and death is certain.
 - In a country, which families will have children this year is random, but on a macro level, the birth rate of the country this year is roughly predictable.
- It is precisely based on this holistic and global thinking framework that probability theory has become the foundation for numerous disciplines.



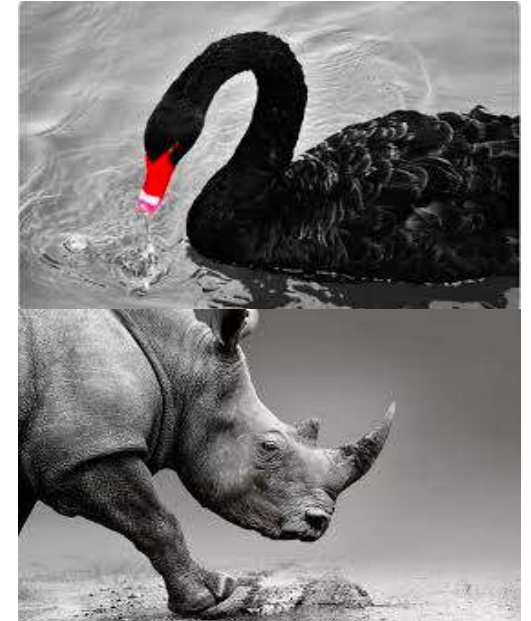
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2.3 Basic Concepts of Probability Theory

- Probability theory studies random events, but what exactly is “random”?
- Mathematicians have debated for many years without reaching a unified definition of “random”. The only consensus is that **randomness suggests unpredictability**.
- Specifically, randomness refers to not knowing which outcome an event might produce, while being aware of all possible outcomes.
- Although randomness implies uncertainty, it is not entirely uncertain.
- A simply understanding of the distinction between “uncertainty” and “randomness” through “**black swan events**” and “**gray rhino events**”:
 - Black swan events: the type of risk is unknown and lies beyond the scope of our understanding. E.g., the terrorist attack on 2001/09/11 and the COVID-19 pandemic.
 - Gray rhino events: the type of risk is known and still within the scope of our understanding. E.g., the Chernobyl explosion in 1986 and the U.S. subprime mortgage crisis in 2007.



看不见的黑天鹅
看得见的灰犀牛

2.3 Basic Concepts of Probability Theory

- Based on the understanding of “randomness”, we introduce some basic concepts

Random Experiment

A **random experiment** (随机试验), often simply called an **experiment**, represents the realization or observation of a random phenomenon and has the following characteristics:

- it can be repeated under the same conditions;
- all possible outcomes are clearly known;
- exactly one of these possible outcomes occurs each time, but it cannot be determined in advance which outcome will occur.

Sample Space and Random Event

- Each possible **fundamental outcome** (基本结果) of a random experiment is called a **sample point** (样本点).
- A set that includes all sample points of the experiment is called the **sample space** (样本空间), usually denoted as Ω .
- From a set theory perspective, a **random event** (随机事件), or **event**, is a subset of the sample space Ω , typically denoted by uppercase letters (e.g., A, B, C , etc.)
- If the outcome is a sample point in event A , then we say that event A happens.



2.3 Basic Concepts of Probability Theory

- With these concepts, we can describe and study random phenomena mathematically.

Example 2.1

A person driving to work needs to pass through three traffic lights. At each traffic light, he either comes to a red light (denoted as 0) or a green light (denoted as 1). Then:

- The sample space is $\Omega = \{000, 001, 010, 100, 011, 101, 110, 111\}$;
- The event that “just one of the three traffic lights is red” is $A = \{011, 101, 110\} \subset \Omega$.

Example 2.2

A couple decides to have children and plans to have children until a boy is born. Write B for boy and G for girl, then:

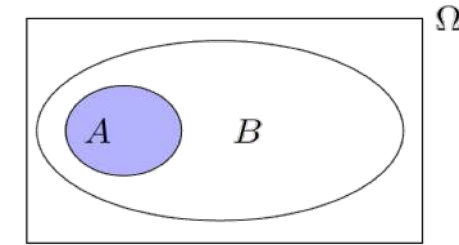
- The sample space is $\Omega = \{B, GB, GGB, GGGB, \dots\}$;
- This example shows that the sample space is not necessarily finite.



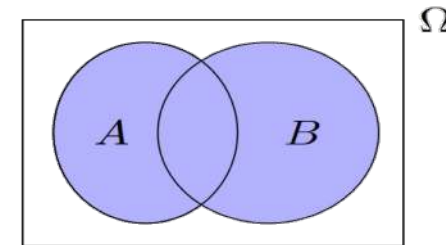
2.3 Basic Concepts of Probability Theory

- From a set theory perspective, the relationships and operations of events become very clear:

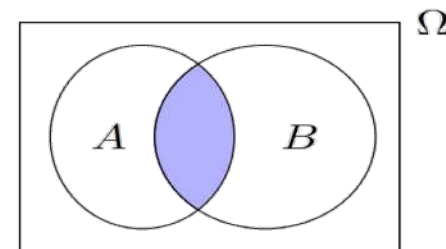
- **Inclusion** $A \subset B$: All sample points in event A are also in B , i.e., B happens when A happens. (包含)



- **Sum/union** $A \cup B (A + B)$: $= \{\omega | \omega \in A \text{ or } \omega \in B\}$, at least one of A and B happens. (和/并)



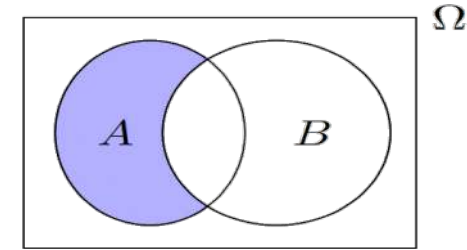
- **Product/intersection** $A \cap B (AB)$: $= \{\omega | \omega \in A \text{ and } \omega \in B\}$, A and B both happen. (积/交)



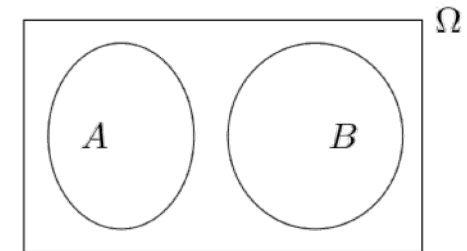
2.3 Basic Concepts of Probability Theory

- From a set theory perspective, the relationships and operations of events become very clear:

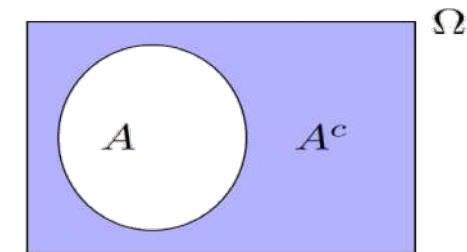
- **Difference $A - B$ ($A \setminus B$):** $= \{\omega | \omega \in A \text{ and } \omega \notin B\}$,
 A happens and B does not happen. (差)



- **Mutually exclusive/disjoint $A \cap B = \emptyset$:** A and B
cannot happen at the same time. (互斥/互不相容)



- **Complement $A \cap B = \emptyset$ & $A \cup B = \Omega$:** either A or B
happens, denoted as $B = A^c$ or $\bar{A}$. (对立/互补)



2.3 Basic Concepts of Probability Theory

- The operations of events obey certain rules similar to the rules of sets:

- **Communicative laws (交换律):**

$$A \cup B = B \cup A, \quad A \cap B = B \cap A$$

- **Associative laws (结合律):**

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

- **Distributive law (分配律):**

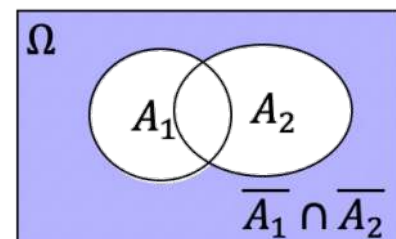
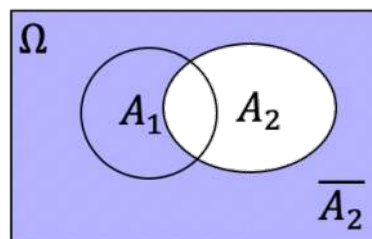
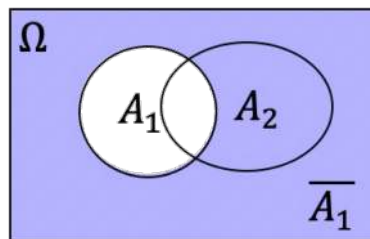
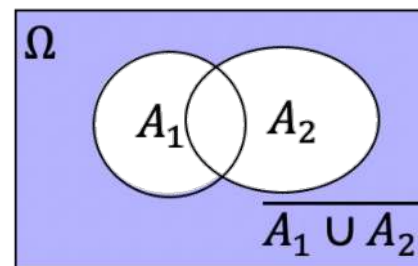
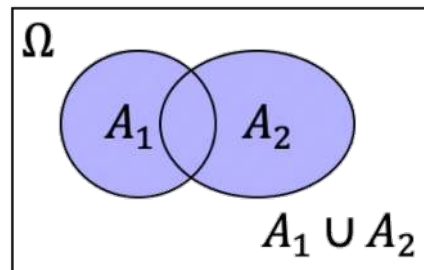
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

- **De Morgan's laws (德·摩根律):**

$$\overline{\bigcup_{i=1}^{\infty} A_i} = \bigcap_{i=1}^{\infty} \bar{A}_i, \quad \overline{\bigcap_{i=1}^{\infty} A_i} = \bigcup_{i=1}^{\infty} \bar{A}_i$$

A brief illustration of the De Morgan's laws



2.3 Basic Concepts of Probability Theory

- Having defined the sample space and random event, we can discuss the probability of events.
- **Question:** What is “probability”?
- The most intuitive understanding of probability: **a quantitative description of the likelihood of a random event occurring.**
- Under this understanding, a natural question is: how is the quantitative description obtained?
 - Before the calculation of probability, let’s explain the idea behind this this quantitative description.
 - Since a random event is a subset of the sample space, **its probability is proportional to its size relative to the sample space.**
- Therefore, calculating probability presupposes the identification of all possible outcomes, **ensuring the completeness of the sample space.**
- The reason black swan events are so daunting is because they are not in our sample space at all.
- In a sense, our exploration of the world is a process of refining and expanding our sample space.



2.3 Basic Concepts of Probability Theory

- The we provide the formal **mathematical definition** of probability.

Probability (simplified version)

Probability measure (概率测度), or simply **probability**, is a real-valued function defined on subsets of the sample space Ω , satisfying the following three axioms:

- **Non-negativity** (非负性): for any event $A \subseteq \Omega$, we have $P(A) \geq 0$.
- **Normalization** (规范性): $P(\Omega) = 1$.
- **Additivity** (可加性): for any **mutually exclusive** events $A_1, A_2, \dots$, we have

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$



2.3 Basic Concepts of Probability Theory

- Following the three axioms, many useful rules for computing probability can be derived:

- $P(\emptyset) = 0$;

- If $A \subseteq B$, then $P(A) \leq P(B)$;

- $P(\overline{A}) = 1 - P(A)$;

- $0 \leq P(A) \leq 1$;

- The general addition rule (加法定律): $P(A \cup B) = P(A) + P(B) - P(AB)$

- Inclusion-exclusion principle(容斥原理):

$$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n) = \sum_{i=1}^n P(A_i) - \sum_{1 \leq i < j \leq n} P(A_i A_j) + \sum_{1 \leq i < j < k \leq n} P(A_i A_j A_k) \\ - \sum_{1 \leq i < j < k < l \leq n} P(A_i A_j A_k A_l) + \dots + (-1)^{n-1} P(A_1 A_2 \dots A_n)$$

+ for odd counts
- for even counts



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2.4 Calculating Probability

- Probability is a precise description of the likelihood of a random event occurring. Is it necessary to precisely compute probability?
- In everyday life, it may not be necessary, as it is difficult for us to distinguish between events with probabilities of 0.3 and 0.4.
- In professional fields, however, precise probability measurement becomes very important. E.g.,
 - In a **casino**, through precise probability measurement and design, the house only needs to have a slightly higher probability of winning than the players to make a profit from an overall perspective.
 - **Insurance companies** also operate a probability-based business. They design and price insurance products by accurately calculating the probability of claims, ensuring that they make a profit overall.



2.4 Calculating Probability

- The probability computations in early days were mostly based on a relatively simple model known as the **classical model of probability** (古典概型).

Classical Model of Probability

If a random experiment satisfies:

- there are only finite number of sample points in the sample space Ω , i.e.,

$$\Omega = \{\omega_1, \omega_2, \dots, \omega_n\},$$

- each sample point is equally likely to occur, i.e.,

$$P(\{\omega_1\}) = P(\{\omega_2\}) = \dots = P(\{\omega_n\}) = \frac{1}{n},$$

then the probability model of the experiment is called a **classical model of probability**.

Probability Computation Under the Classical Model of Probability

Assume that there are k sample points in event A , then the probability of event A is

$$P(A) = \frac{\text{Number of sample points in } A}{\text{Total number of sample points in } \Omega} = \frac{k}{n}.$$

Hence, under the classical model of probability, probability computation only involve counting the number of sample points in the sample space and the event of interest.

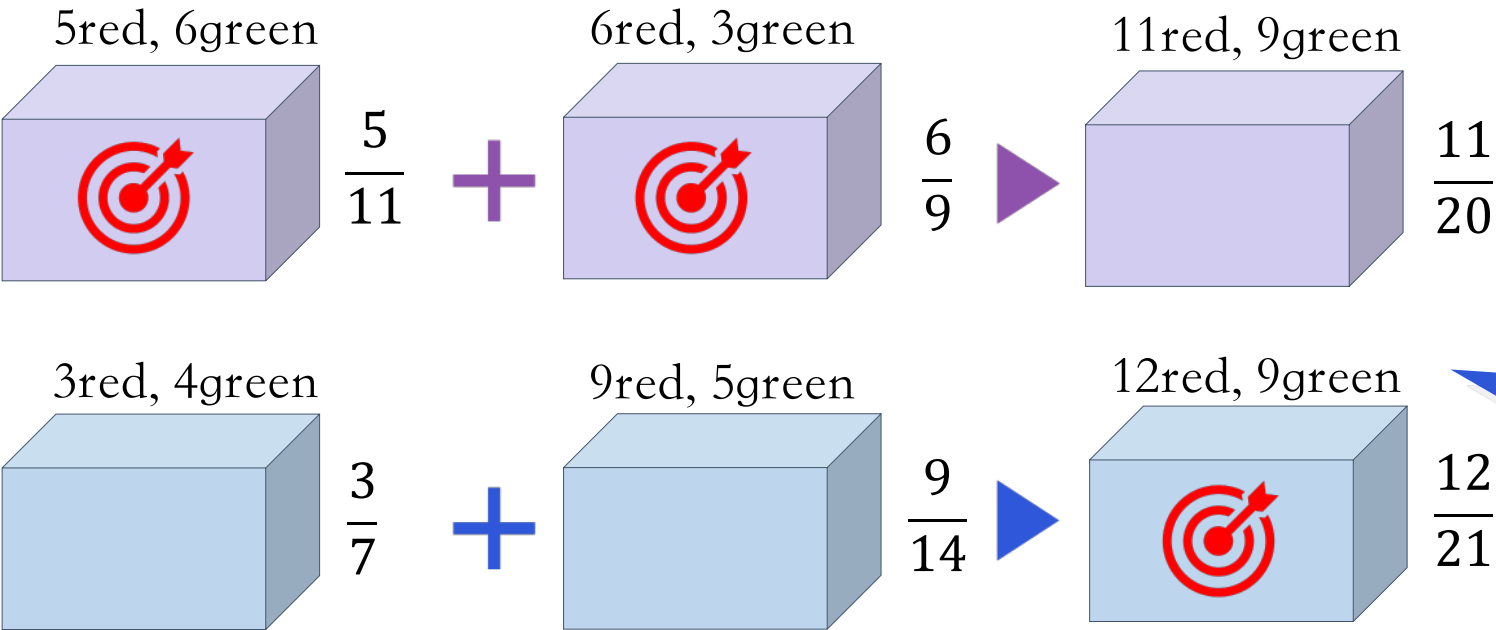
- The probability of a random event is the proportion of that event, a subset of the sample space, within the sample space.



2.4 Calculating Probability

Example 2.3

If two boxes each contain a certain number of **red** and **green** balls, and you are allowed to first choose a box and then randomly pick a ball from the chosen box. If a **red** ball is picked, you win a prize. Then which box should you choose from?



(**Real-world example:** UC Berkeley admissions data from 1973)



2.4 Calculating Probability

- However, more often, the number of sample points is not easy to determine and requires more systematic counting methods.
- We introduce/review two commonly used counting methods: the addition and multiplication principles, and permutations and combinations.

Addition and Multiplication Principles

- **Addition principle (加法原理):** If there are n types of methods to complete a task, with m_1 specific methods in the first type, m_2 specific methods in the second type, ..., and m_n specific methods in the n th type, then the total number of specific methods to complete this task is:

$$N = m_1 + m_2 + \cdots + m_n.$$

- **Multiplication principle (乘法原理):** If there are n steps to complete a task, with m_1 possible methods for the first step, m_2 possible methods for the second step, $\cdots$, m_n methods for the n th step, then the total number of methods to complete this task is:

$$N = m_1 \times m_2 \times \cdots \times m_n.$$



2.4 Calculating Probability

Permutation and Combination

- **Permutation (排列)**: If k elements are randomly selected **without replacement (无重复随机抽取)** from n distinct elements ($k \leq n$) and **placed in order**, then the number of different permutations is

$$A_n^k = n(n-1) \dots (n-k+1) = \frac{n!}{(n-k)!}.$$

- **Combination (组合)**: The number of combinations of k elements randomly selected **without replacement (无重复随机抽取)** from n distinct elements ($k \leq n$), where the **order does not matter**, is given by

$$C_n^k = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

- **Think**: what's the permutation and combination under the case of **randomly selecting with replacement (有重复随机抽取)**?



2.4 Calculating Probability

Example 2.4

Suppose a class of n students has been allocated m ($< n$) concert tickets by the university. The teacher decides to distribute the tickets by drawing lots.

The teacher prepares a hat containing n slips of paper, with m slips marked with a "1" and the remaining slips marked with a "0".

The students take turns drawing slips from the hat, and those who draw a slip marked with a "1" will get a concert ticket.

If you are one of the students in the class and you really want to get a ticket, would you choose to draw early or late?



2.4 Calculating Probability

Solution

- The essential question is: Suppose you are the l th to draw a slip, does the probability of you getting a ticket depend on l ?
- To compute probability, we first need to figure out what's the sample space?
- Considering the ticket drawing of the class as a random experiment, then each possible outcome of the drawing process is a sample point (i.e., an n -digit number with m digits being 1 and the remaining being 0), the sample space is the set of all these sample points.
- Then, determine the number of sample points in the sample space: C_n^m .
- Let A denote the event that you get a ticket, then the number of sample points in event A is: C_{n-1}^{m-1} .
- Put these together, the probability of event A is

$$P(A) = \frac{\text{Number of sample points in } A}{\text{Total number of sample points in } \Omega} = \frac{C_{n-1}^{m-1}}{C_n^m} = \frac{m}{n}.$$

- Therefore, the probability of getting a ticket **does not** depend on the order in which you draw a slip!



2.4 Calculating Probability

Example 2.5

(**Birthday problem**) There are n ($n < 365$) students in a class, what's the probability that at least two students have the same birthday? (leap years are not taken into account)

Solution

- Each sequence of n birthdays is a sample point, and the sample space is the set of all possible sequences of n birthdays.
- Then, the number of sample points in the sample space is: 365^n .
- Let A be the event that at least two students have the same birthday, then $\bar{A}$ is the event that every student has a different birthday.
- It is easier to determine the number of sample points in $\bar{A}$:

$$A_{365}^n = 365 \times 364 \times \cdots \times (365 - n + 1)$$

- Therefore, the probability of event A is computed as:

$$P(A) = 1 - P(\bar{A}) = 1 - \frac{A_{365}^n}{365^n}.$$



2.4 Calculating Probability

Solution



Results							
n	20	25	30	40	50	55	100
$P(A)$	0.41	0.57	0.71	0.89	0.97	0.99	0.9999997

- In a class of 50 students, the probability that at least two people share the same birthday is as high as 97%.
- In a class of 100 students, it is almost certain that at least two people will share the same birthday.
- Does this surprise you?



2.4 Calculating Probability

Example 2.6

(Matching problem 配对问题)

- There are n students in a class with student IDs $1, 2, \dots, n$.
- Before the next Children's Day, everyone will prepare a gift which will be numbered as his/her student ID.
- All the gifts will be put into a bag, and everyone will randomly select a gift from the bag.
- What's the probability that at least one student get his/her own gift?



Solution

- Each permutation of $\{1, 2, \dots, n\}$ is a sample point, representing the gift numbers corresponding to students with IDs $1, 2, \dots, n$.
- So, the number of sample points in the sample space is: $A_n^n = n!$.
- Let A be the event that at least one student get his/her own gift, then directly determining the number of sample points in A is not so straightforward.



2.4 Calculating Probability

Solution

- The idea is to break down vague and complex problems into clear and simple ones.
- Let A_i be the event that the student with ID i get his/her own gift, then

$$\begin{aligned} P(A) &= P(A_1 \cup A_2 \cup \dots \cup A_n) \\ &= \sum_{i=1}^n P(A_i) - \sum_{i < j} P(A_i A_j) + \dots + (-1)^{n-1} P(A_1 A_2 \dots A_n) \end{aligned}$$

- The events $A_i, A_i A_j, A_1 A_2 \dots A_n$ are very clear and their probability computation is simple:

$$P(A_i) = \frac{(n-1)!}{n!} = \frac{1}{C_n^1}, \quad P(A_i A_j) = \frac{(n-2)!}{n!} = \frac{1}{2! C_n^2}, \dots$$

- Put these together, we have:

$$\begin{aligned} P(A) &= \frac{1}{C_n^1} \times C_n^1 - \frac{1}{2! C_n^2} \times C_n^2 + \frac{1}{3! C_n^3} \times C_n^3 + \dots + (-1)^{n-1} \frac{1}{n!} \\ &= 1 - \frac{1}{2!} + \frac{1}{3!} + \dots + (-1)^{n-1} \frac{1}{n!} \approx 1 - e^{-1} \approx 0.632. \end{aligned}$$



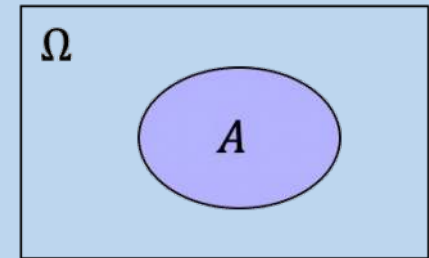
2.4 Calculating Probability

- The classical model of probability assumes finite number of sample points and equal likelihood.
- Another model, called **the geometric model of probability (几何概型)**, is an extension of the classical model to an infinite number of sample points while maintaining equal likelihood.

Geometric Model of Probability and its Probability Computation

If a random experiment can be represented as randomly throwing a point onto a bounded region Ω , where the point is **equally likely** to land at any position within the region, then the probability model of the experiment is called a **geometric model of probability**. Let A be the event that the point lands at a subregion A of Ω , then the probability of event A is computed as

$$P(A) = \frac{\text{The length/area/volume of } A}{\text{Total length/area/volume of } \Omega}.$$



- No matter which model is used, probability is essentially the proportion of the random event, a subsect of the sample space, within the sample space.



2.4 Calculating Probability

Example 2.7

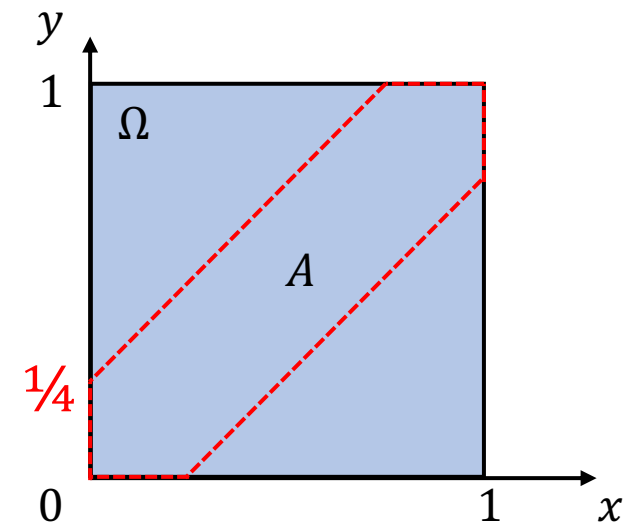
- Romeo and Juliet plan to meet in an evening, and each will arrive at the garden between 7pm and 8pm, with all pairs of arrival time equally likely.
- The first to arrive will wait for 15 minutes and will leave if the other has not yet arrived.
- What is the probability that they will meet?



Solution

- Let 7pm be the origin, x -axis and y -axis be the arrival time of Romeo and Juliet, respectively.
- Then the sample space is $\Omega = \{(x, y) \in [0, 1] \times [0, 1]\}$.
- For Romeo and Juliet to meet, we need $|x - y| \leq 1/4, 0 \leq x, y \leq 1$.

$$P(A) = \frac{\text{The area of } A}{\text{Total area of } \Omega} = 1 - \frac{3}{4} \times \frac{3}{4} = \frac{7}{16}.$$



Chapter 2 Understanding Probability

- 2.1 Determinism and Randomness
- 2.2 Development of Probability Theory
- 2.3 Basic Concepts of Probability Theory
- 2.4 Calculating Probability
- 2.5 Conditional Probability and Independence

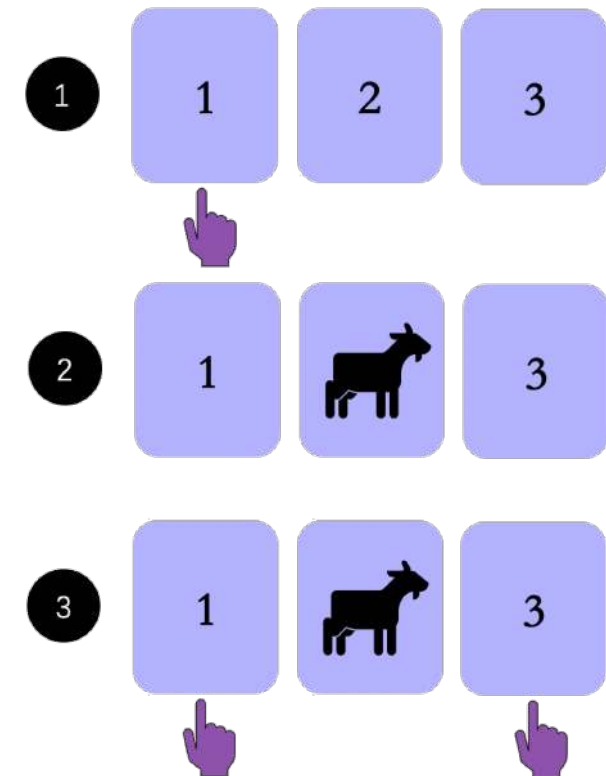


2.5 Conditional Probability and Independence

- Let's first look at a problem that once sparked widespread discussion in the United States—the **Monty Hall problem (三门问题)**.

Example 2.8

- The problem originates from the American TV show *Let's Make a Deal*, named after the show's host, Monty Hall.
- The show came into the public eye in 1975. Contestants would see three closed doors, with a car behind one of them and a goat behind each of the other two. If they successfully pick the door with the car, they will win the car.
- You are first asked to choose one of the doors, say Door 1.
- Then, the host (who knows where the car is) opens one of the remaining two doors, revealing a goat, say behind Door 2.
- Finally, the host gives you a chance to change your choice—either stick with your original choice (Door 1) or switch to Door 3.
- What's your choice?



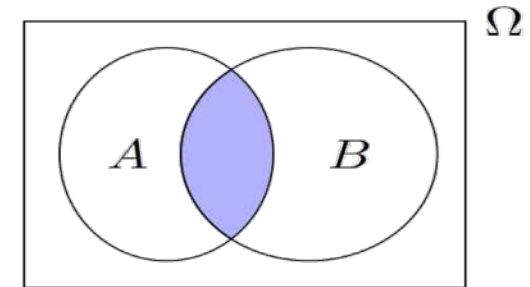
2.5 Conditional Probability and Independence

- After learning about conditional probability, you will find a logically rigorous way to solve the Monty Hall problem.
- Conditional probability refers to the idea that the probability of a random event will change given the occurrence of another event.

Conditional Probability

Let A and B be two random events and $P(B) > 0$. Then the conditional probability (条件概率) of event A given that event B occurs is defined as

$$P(A|B) \triangleq \frac{P(A \cap B)}{P(B)} = \frac{P(AB)}{P(B)}.$$



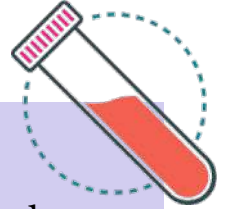
- The intuitive understanding of $P(A|B)$ is that the occurrence of event B provides new information that updates our belief on the likelihood of event A .
- The idea behind this definition is that if event B has occurred, the sample space becomes B instead of Ω .



2.5 Conditional Probability and Independence

Example 2.9

- You have a blood test for a rare disease that occurs by chance in 1 person in 100,000.
- If you have the disease, the test will report that you do with probability 0.95 (and that you do not with probability 0.05).
- If you do not have the disease, the test will report a false positive with probability 0.001.
- If the test says you have the disease, what is the probability that you actually have the disease?



Thinking

- First, we express the given conditions in the example using probability language.
- Let A be the event that the test reports a positive result for a randomly chosen person, B be the event that a randomly chosen person has the disease, then:

$$P(A|B) = 0.95, P(A|\bar{B}) = 0.001, P(B) = 0.00001.$$

- The probability to be computed is expressed as $P(B|A) = P(AB)/P(A)$.
- How to compute $P(AB)$ and $P(A)$ based on the given conditions?

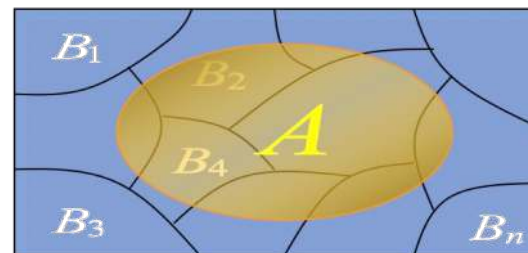


2.5 Conditional Probability and Independence

- To solve the problem, we introduce the **multiplication law** (乘法定律, used to compute $P(AB)$) and the **law of total probability** (全概率定律, used to compute $P(A)$).

Multiplication Law

Let A and B be two random events and $P(B) > 0$. Then

$$P(AB) = P(A|B)P(B).$$


Law of Total Probability

Let A and B be two random events, then (assume that $P(A|B) = 0$ if $P(B) = 0$)

$$P(A) = P(AB) + P(A\bar{B}) = P(A|B)P(B) + P(A|\bar{B})P(\bar{B}).$$

More generally, if $B_1, B_2, \dots, B_n$ are n **mutually exclusive** random events, and $B_1 \cup B_2 \cup \dots \cup B_n = \Omega$, then we call $B_1, B_2, \dots, B_n$ a **partition** (分割/分划) of the sample space Ω , and

$$\begin{aligned} P(A) &= P(A \cap \Omega) = P(AB_1 \cup AB_2 \cup \dots \cup AB_n) = P(AB_1) + P(AB_2) + \dots + P(AB_n) \\ &= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n). \end{aligned}$$



2.5 Conditional Probability and Independence

Solution of Example 2.9

- By the multiplicative law:

$$P(AB) = P(A|B)P(B) = 0.95 \times 0.00001.$$

- By the law of total probability:

$$P(A) = P(A|B)P(B) + P(A|\bar{B})P(\bar{B}) = 0.95 \times 0.00001 + 0.001 \times 0.99999.$$

- Therefore, we have

$$P(B|A) = \frac{P(AB)}{P(A)} = \frac{0.95 \times 0.00001}{0.95 \times 0.00001 + 0.001 \times 0.99999} \approx 0.0094.$$

- Even if the test accuracy is very high, the probability that a positive result is correct is very small.
- Does this contradict your experience or intuition? Why does this happen?
- The reason is that incidence of the disease in the population is very low.



2.5 Conditional Probability and Independence

- Actually, a famous theorem has been used in the solution of the blood test example, which is the Bayes' theorem (贝叶斯定理).

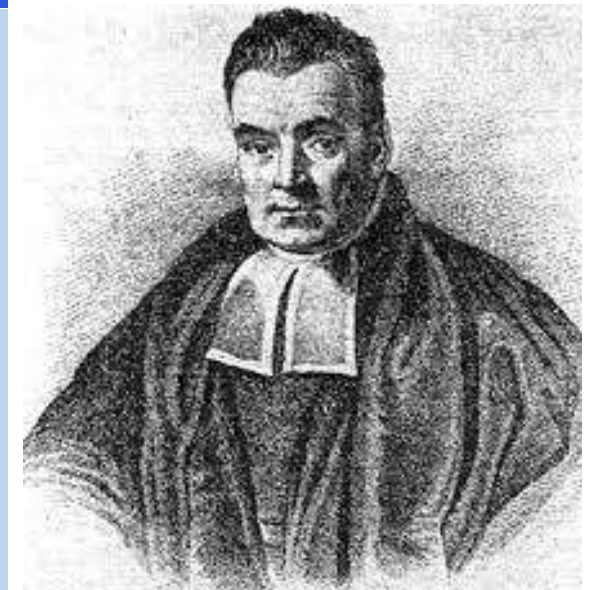
Bayes' Theorem / Bayes' Rule

Let $B_1, B_2, \dots, B_n$ be random events and $B_1, B_2, \dots, B_n$ is a partition of the sample space. Then for any event A such that $P(A) > 0$, we have

$$P(B_i|A) = \frac{P(B_i)P(A|B_i)}{P(A)} = \frac{P(B_i)P(A|B_i)}{\sum_{i=1}^n P(B_i)P(A|B_i)}.$$

- $P(B_i|A)$ is the probability of B_i given that A occurs, also termed the **posterior probability** (后验概率) of B_i .
 - $P(B_i)$ is called the **prior probability** (先验概率) or marginal probability (边缘概率) of B_i , which refers to the probability value in the absence of any other prior information.
- The Bayes' rule is widely used across various fields, especially in scenarios involving uncertainty and decision-making under incomplete information.

Thomas Bayes
(1701-1761)



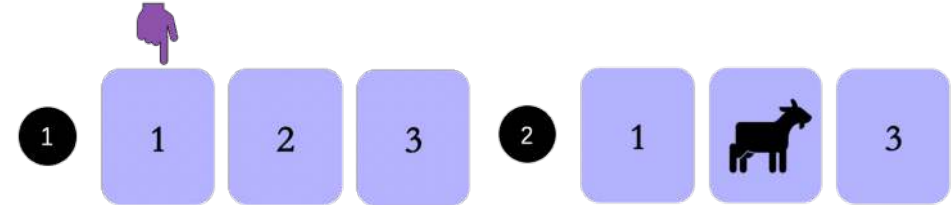
2.5 Conditional Probability and Independence

- A real-world example of the application of Bayes' rule – the **Clinical Decision Support System**
 - $B_1, B_2, \dots, B_n$ represent various diseases, and A represents a certain symptom or indicator (such as the level of transaminase in the blood).
 - The prior probability $P(B_i)$ can be determined using statistical methods.
 - $P(A|B_i)$ can be determined using medical knowledge.
 - By applying the Bayes' rule, we can calculate the posterior probability $P(B_i|A)$.
 - The diseases corresponding to the larger $P(B_i|A)$ can be provided to the doctor for further clinical diagnosis.



2.5 Conditional Probability and Independence

- Lastly, let's solve the Monty Hall problem.



Solution of Example 2.8

- Without loss of generality, suppose that you chose Door 1 and the host opened Door 2.
- Let B_1, B_2, B_3 be event that there is a car behind Door 1, 2, 3 and A be the event that the host opens Door 2. Then:

$$P(B_1) = P(B_2) = P(B_3) = \frac{1}{3}, P(A|B_1) = \frac{1}{2}, P(A|B_2) = 0, P(A|B_3) = 1.$$

- The probability of interest is $P(B_1|A)$ and $P(B_3|A)$. By the law of total probability, we have

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) = \frac{1}{2} \times \frac{1}{3} + 0 \times \frac{1}{3} + 1 \times \frac{1}{3} = \frac{1}{2}.$$

- Finally, by the Bayes' rule:

$$P(B_1|A) = \frac{P(A|B_1)P(B_1)}{P(A)} = \frac{1/2 * 1/3}{1/2} = \frac{1}{3}, \quad P(B_3|A) = \frac{P(A|B_3)P(B_3)}{P(A)} = \frac{1 * 1/3}{1/2} = \frac{2}{3}.$$

- Therefore, you should switch to Door 3!



2.5 Conditional Probability and Independence

- Finally, let's discuss the **independence** (独立性) between events. You might have learned about some definitions of independence before, like:
 - The independence between two events means that the occurrence of one event does not affect the probability of another event.
 - If $P(AB) = P(A)P(B)$, then we claim that events A and B are independent.
- However, the first is not a mathematically rigorous definition, the second seems difficult to understand intuitively.
- With the help of conditional probability, the independence between events can be easily and clearly defined.
- “The occurrence of A does not affect the probability of B , and vice versa” can be expressed as

$$P(A|B) = P(A), P(B|A) = P(B).$$

- By the multiplication law, we therefore have

$$P(AB) = P(A|B)P(B) = P(B|A)P(A) = P(A)P(B).$$



2.5 Conditional Probability and Independence

- In mathematics, it is not difficult to determine if events are independent as it is clearly defined.
- However, in real life, it may not be easy to discern whether events are independent. E.g., the butterfly effect:
 - “the flap of a butterfly’s wings in South America could ultimately cause a tornado in Texas.”
 - So, something seemingly unrelated could bring about a significant change.
- Many instances of independence in real life are assumptions we make to greatly simplify problems, i.e., **independence is often just a mathematical model we use to describe random events.**

Example 2.10

Both Taobao’s shopping system and the 12306 railway ticketing system face situations where a huge number of users use the service at the same time, but for backend developers, the development difficulty of the two systems is different. Which one is more difficult?



